

RETAKE EXAM

This is a 2 hours exam. No documents allowed

I – RISK SHARING (50% of the points)

We consider the setup of Franck Portier's lecture 3 "Asset Pricing in General Equilibrium". Households are indexed by $i = 1, \dots, I$. Each owns a stochastic endowment of one good $y_t^i = y^i(s_t)$, and s_t is publicly observable. Each household purchase a history-dependant consumption plan $c^i = \{c_t^i(s^t)\}_{t=0}^\infty$. Household i objective is to maximize

$$U(c^i) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t u[c_t^i(s^t)] \pi(s^t | s_0) = E_0 \sum_{t=0}^{\infty} \beta^t u[c_t^i(s^t)]$$

$\pi(s^t | s)$ is a Markov chain with initial distribution $\pi_0(s)$. Trading occurs after s_0 has been observed.

Household trade a full set of dated state-contingent claims to consumption. $q_t^0(s^t)$ is the price of a claim on time- t consumption, contingent on history s^t , in terms of a numéraire not specified.

1 – Write the budget constraint of household i .

$$\sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) c_t^i(s^t) = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) y^i(s_t)$$

2 – Show that optimality of choices for all the households implies that marginal utilities of consumption in any given state are proportional among households.

Let μ^i be the Lagrange multiplier of this constraint, FOC:

$$\frac{\partial U(c^i)}{\partial c_t^i(s^t)} = \mu^i q_t^0(s^t)$$

We have

$$\frac{\partial U(c^i)}{\partial c_t^i(s^t)} = \beta^t u'[c_t^i(s^t)] \pi(s^t | s_0)$$

so that

$$\beta^t u'[c_t^i(s^t)] \pi(s^t | s_0) = \mu^i q_t^0(s^t)$$

This equation implies

$$\frac{u'[c_t^i(s^t)]}{u'[c_t^j(s^t)]} = \frac{\mu^i}{\mu^j} \quad (\star)$$

which means that ratios of marginal utilities between pairs of agents are constant across all states and dates.

3 – Define a price system, an allocation, a feasible allocation, a competitive equilibrium

A **price system** is a sequence of functions $\{q_t^0(s^t)\}_{t=0}^\infty$. An **allocation** is a list of sequences of functions $\{c_t^i(s^t)\}_{t=0}^\infty$, one for each i . A **feasible allocation** satisfies

$$\sum_i y^i(s_t) \geq \sum_i c_t^i(s^t)$$

A **competitive equilibrium** is a feasible allocation and price system such that the allocation solves each household problem.

4 – Using households first order conditions and the feasibility constraint, show that the competitive equilibrium allocation is not history dependent – i.e. $c_t^i(s^t) = \bar{c}^i(s_t)$.

Note that $(\star)$ implies

$$c_t^i(s^t) = u'^{-1} \left\{ u'[c_t^1(s^t)] \frac{\mu^i}{\mu^1} \right\} \quad (\star\star)$$

Substituting into feasibility condition at equality gives

$$\sum_i u'^{-1} \left\{ u'[c_t^1(s^t)] \frac{\mu^i}{\mu^1} \right\} = \sum_i y^i(s_t)$$

The RHS of this equation does not depend on the entire history s^t , but only on current state s_t , therefore the LHS, therefore $c_t^1(s^t)$. Then, from $(\star\star)$, it is also the case for all $c_t^i(s^t)$. One then has the proposition: “The competitive equilibrium allocation is not history dependent; $c_t^i(s^t) = \bar{c}^i(s_t)$ ”

5 – Assuming that preferences are CRRA, $u(c) = (1 - \gamma)^{-1} c^{1-\gamma}$, $\gamma > 0$, show that individual consumption is perfectly correlated with the aggregate endowment or aggregate consumption.

$(\star)$ implies

$$c_t^i = c_t^j \left(\frac{\mu^i}{\mu^j} \right)^{-\frac{1}{\gamma}}$$

The time- t elements of consumption allocations to distinct agents are constant fractions of one another. The individual consumption is perfectly correlated with the aggregate endowment or aggregate consumption. The fractions assigned to each individual are independent of the realization of s_t . There is extensive cross-time cross-state consumption smoothing.

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Assume now that preferences are CARA: $u(c^i(s^t), \delta^i(s^t)) = -\frac{1}{\sigma} e^{-\sigma(c^i - \delta^i)}$, with δ_t^i being a idiosyncratic preference shock. Also assume that endowments are given by $y^i(s^t) = \bar{y}_t^i + \eta^i(s^t) + \varepsilon^i(s^t)$ where $\bar{y}_t^i$ is a deterministic component, $\eta^i(s^t)$ is an aggregate shock to individual i 's endowment (aggregate in the sense that $\sum_{i=1}^I \eta^i(s^t) = \eta^a(s^t)$, but magnitude varies across i) and $\varepsilon^i(s^t)$ is an idiosyncratic shock (in the sense that $\sum_{i=1}^I \varepsilon^i(s^t) = 0$).

Let us consider a planner that maximizes $\sum_{i=1}^I \omega^i \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi(s^t) u(c^i(s^t), \delta^i(s^t))$ s.t. $\sum_{i=1}^I c^i(s^t) \leq \sum_{i=1}^I y^i(s^t)$, where the ω^i are given weights.

6 – Write the first order condition of this problem with respect to $c^i(s^t)$.

FOC for individual i writes (I suppress state notation)

$$\hat{\mu}_t = \frac{\mu_t}{\beta^t \pi_t} = \omega^i \frac{\partial u(c_t^i, \delta_t^i)}{\partial c_t^i} \quad (1)$$

where μ_t is the Lagrange multiplier associated to the resource constraint.

7 – Show that changes in consumption, net of preference shocks, are equalized across individuals. What does it suggest as a possible way to test for full risk sharing?

(1) writes $\hat{\mu}_t = \omega^i e^{-\sigma(c_t^i - \delta_t^i)}$.

Take logs: $c^i(t) = \frac{1}{\sigma} \log \omega^i + \delta_t^i - \frac{1}{\sigma} \log \hat{\mu}_t$ (2)

Aggregate over i : $c_t^a = \frac{1}{\sigma} \log \omega^a + \delta_t^a - \frac{1}{\sigma} \log \hat{\mu}_t$ (3)

with $\log \omega^a = \frac{1}{I} \sum_{i=1}^I \log \omega^i$, $c_t^a = \frac{1}{I} \sum_{i=1}^I c_t^i$, $\delta_t^a = \frac{1}{I} \sum_{i=1}^I \delta_t^i$.

(2) - (3) : $c_t^i = c_t^a + \frac{1}{\sigma} (\log \omega^i - \log \omega^a) + (\delta_t^i - \delta_t^a)$ (4)

Discussion of (4):

- Consumption for individual i is above the economywide average of consumption if the sign of $\log \omega^i - \log \omega^a$ is positive – i.e. for individuals with above-average wealth.
- Idiosyncratic shocks do not show up: perfect risk-sharing.

- Individual consumption varies positively with aggregate consumption

Take first difference of (4):

$$c_{t+1}^i - c_t^i = c_{t+1}^a - c_t^a + [(\delta_{t+1}^i - \delta_t^i) - (\delta_{t+1}^a - \delta_t^a)]$$

- Changes in consumption, net of preference shocks, are equalized across individuals.
- The individual fixed effect in equation (6) is removed with the first difference.

For exponential preferences, we therefore have:

$$\Delta c_t^i = \Delta c_t^a + (\Delta \delta_t^i - \Delta \delta_t^a) \quad (5)$$

Formally, one can estimate

$$\Delta c_t^i = \beta_1 \Delta c_t^a + \beta_2 \Delta y_t^i + u_t^i \quad (6)$$

- u_t^i stands for time-varying individual and aggregate preference shock, measurement error in c , in y .
- predictions of the risk sharing model are $\beta_1 = 1$, $\beta_2 = 0$.
- Note that remove individual fixed effect, this equation is in first difference.

II – EXAM QUESTIONS (*50% of the points*)

Please discuss one of the two statements below, making explicit use of the models and results that we have discussed in this course.

1 – “It is not a good idea to use taxes on capital income to raise government revenue.”

2 – “Business cycle fluctuations are unnecessarily amplified by frictions in macroeconomic adjof prices, investment and economic activity. Government policy should play an active role in reducing these inefficiencies and stabilizing output fluctuations.”